

# Skills Summary

## One-Step Mass and Liquid Volume Problems

*Use this reference while completing your worksheet or when studying.*

---

**The units.** Mass is how heavy something is: grams (g) and kilograms (kg). Liquid volume is how much liquid fills a container: millilitres (mL) and litres (L).

**Choosing the one step.** “In all”, “altogether”, “put together” → **add**. “Left”, “poured out”, “taken away”, “how much more” → **subtract**. Equal groups → **multiply**. Shared equally → **divide**.

**The unit rule.** The answer to a measurement question always carries a unit, and it is the same unit the givens were measured in.

### 1. Mass problems (grams and kilograms)

---

Write both masses down with their unit before you choose an operation. Masses combine exactly like plain numbers — what the unit does is tell you the answer is still a mass.

**Example:** A tin of beans has a mass of 320 g. A tin of corn has a mass of 180 g. What is the mass of both tins together?

**Step 1.** “Together” means the two masses are put into one amount, so this is one addition.

**Step 2.**  $320 + 180$ : make a friendly number first,  $320 + 80 = 400$ , then add the 100 that is left. The two tins have a mass of **500 g**.

**Try it:** A melon has a mass of 460 g and a pear has a mass of 140 g. What is their total mass, in grams?

check: 009 g

### 2. Liquid volume (millilitres and litres)

---

Liquid poured *out* of a container makes the amount smaller, so the answer must be *less* than what you started with. That size check catches most wrong answers before the grader does.

**Example:** A bottle holds 900 mL of water. Sam pours 250 mL into a glass. How much water is left in the bottle?

**Step 1.** Water leaves the bottle, so the one step is a subtraction, and the answer has to come out smaller than 900 mL.

**Step 2.**  $900 - 250$ : take away 200 to get 700, then take away 50. The bottle still holds **650 mL**.

**Try it:** A jug holds 750 mL of lemonade. Ada pours out 400 mL. How much lemonade is left, in millilitres?

check: 050 mL

### 3. Read the measurement, then take one step

---

When the measurements sit in a table, the first job is *reading*: find the two rows the question names and ignore the rest. Then take the single step the question asks for.

**Example:** A record table gives bag A a mass of 700 g and bag B a mass of 450 g. How much heavier is bag A than bag B?

**Step 1.** “How much heavier” compares two masses, so read those two rows and subtract the smaller from the larger.

**Step 2.**  $700 - 450 = 250$ . Bag A is **250** g heavier than bag B.

**Try it:** The same table gives jug A 600 mL of water and jug B 375 mL. How much more water is in jug A, in millilitres?

check: 225 mL

(!) **Watch out:** an answer bigger than what you started with is a warning sign on a “poured out” or “taken away” problem — it usually means the two numbers were added. And never drop the unit: 450 on its own does not say whether the water is 450 mL or the melon is 450 g.